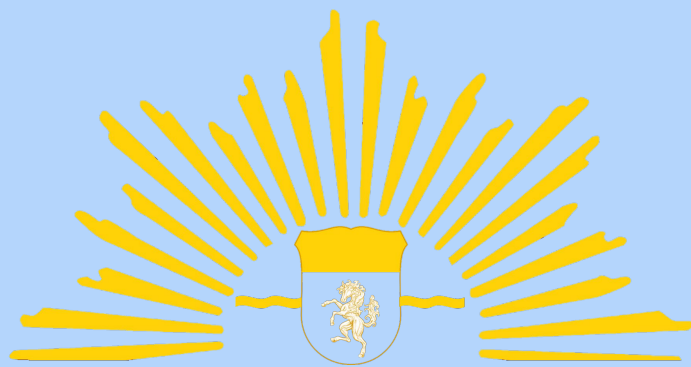


An underwater scene illustration in shades of blue. It features various marine life elements like fish, coral, and seaweed. Scattered throughout are pieces of marine debris: a white pill bottle with a blue cross, a blue surgical mask, and a white glove. The background has wavy lines representing the ocean floor and water layers.

JMT 2022 HeliosX

ENDING AND RECOVERING
FROM MARINE DEBRIS



HeliosX

NVS Praseeth Sashank

Ramanathan AS

Nitya Kairavi V

Riddhima Prasad AS



The Problem



Increase In Marine Debris

Increase in marine debris:

- Adversely affects the marine ecosystem
- Disrupts economies and industries concerned with the seas and oceans
- Contributes to climate change
- Directly impacts human health via food chain
- Etc.



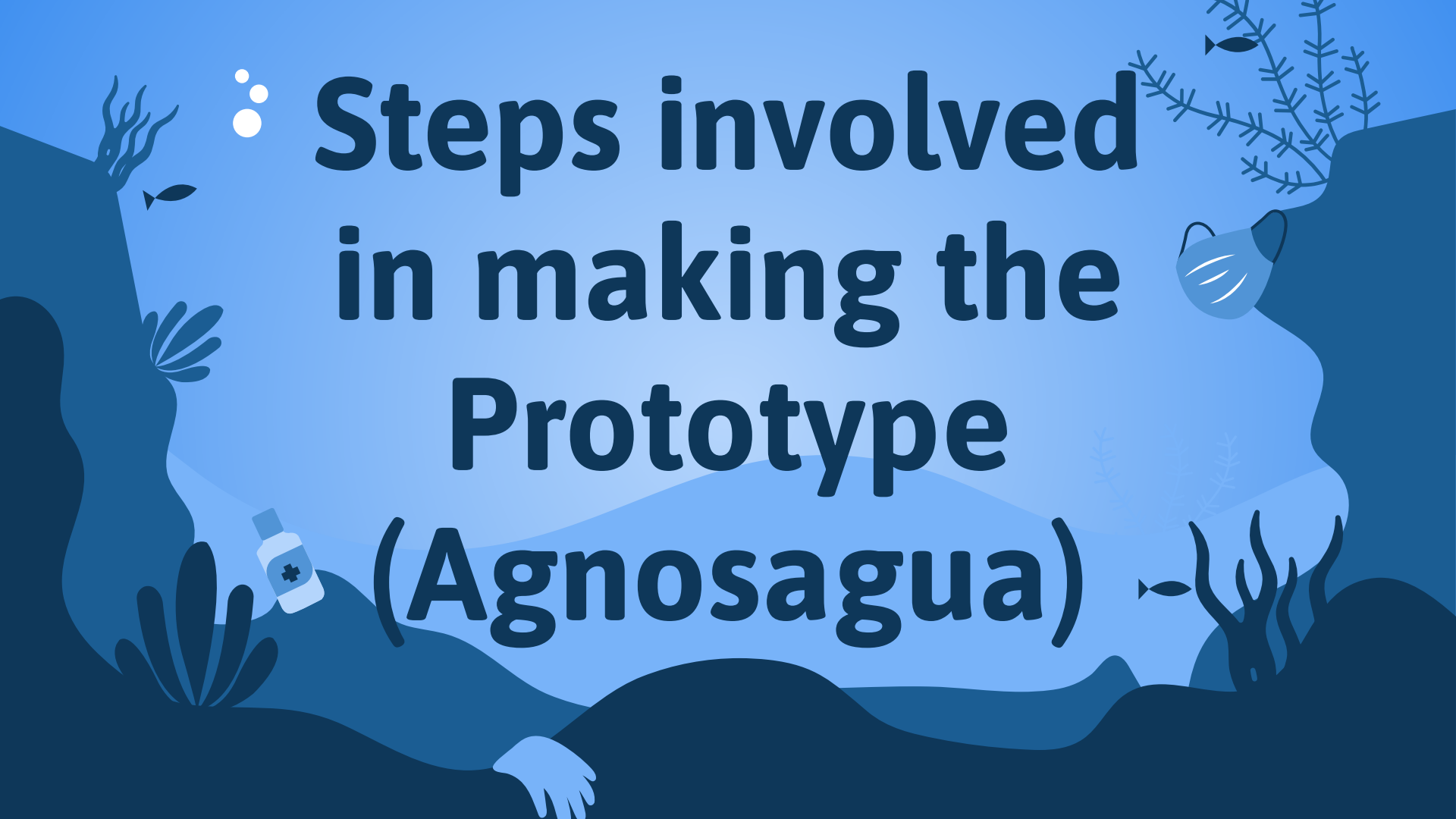
Our Solution

An underwater-themed illustration in various shades of blue. The background features wavy lines representing the ocean floor. On the left, there is a dark blue rock formation with a white medical bottle marked with a blue cross, a small fish, and some coral. On the right, another dark blue rock formation features a white surgical mask, a small fish, and some coral. In the upper left, there are three white bubbles. The overall style is clean and modern.

Agnosagua & Apokathisto

Advanced, eco-friendly, economic ocean rapid-cleaning buoy and segregation and sending of retrieved wastes to recyclers/decontaminators.

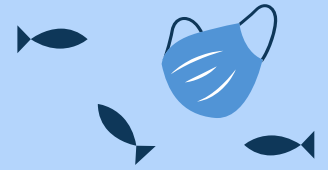


The background is a stylized underwater scene in various shades of blue. It features silhouettes of coral reefs, seaweed, and small fish. A white medical bottle with a blue cross is on the left, and a blue surgical mask is on the right. A hand is visible at the bottom center.

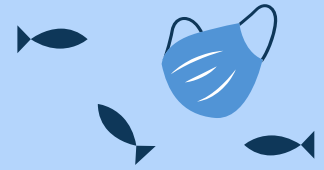
Steps involved in making the Prototype (Agnosagua)

COMPONENTS TO BE MADE:

- Tesla valve
- Fine Sieve
- Torque promoter
- Converging-diverging nozzle
- Electrostatic precipitator



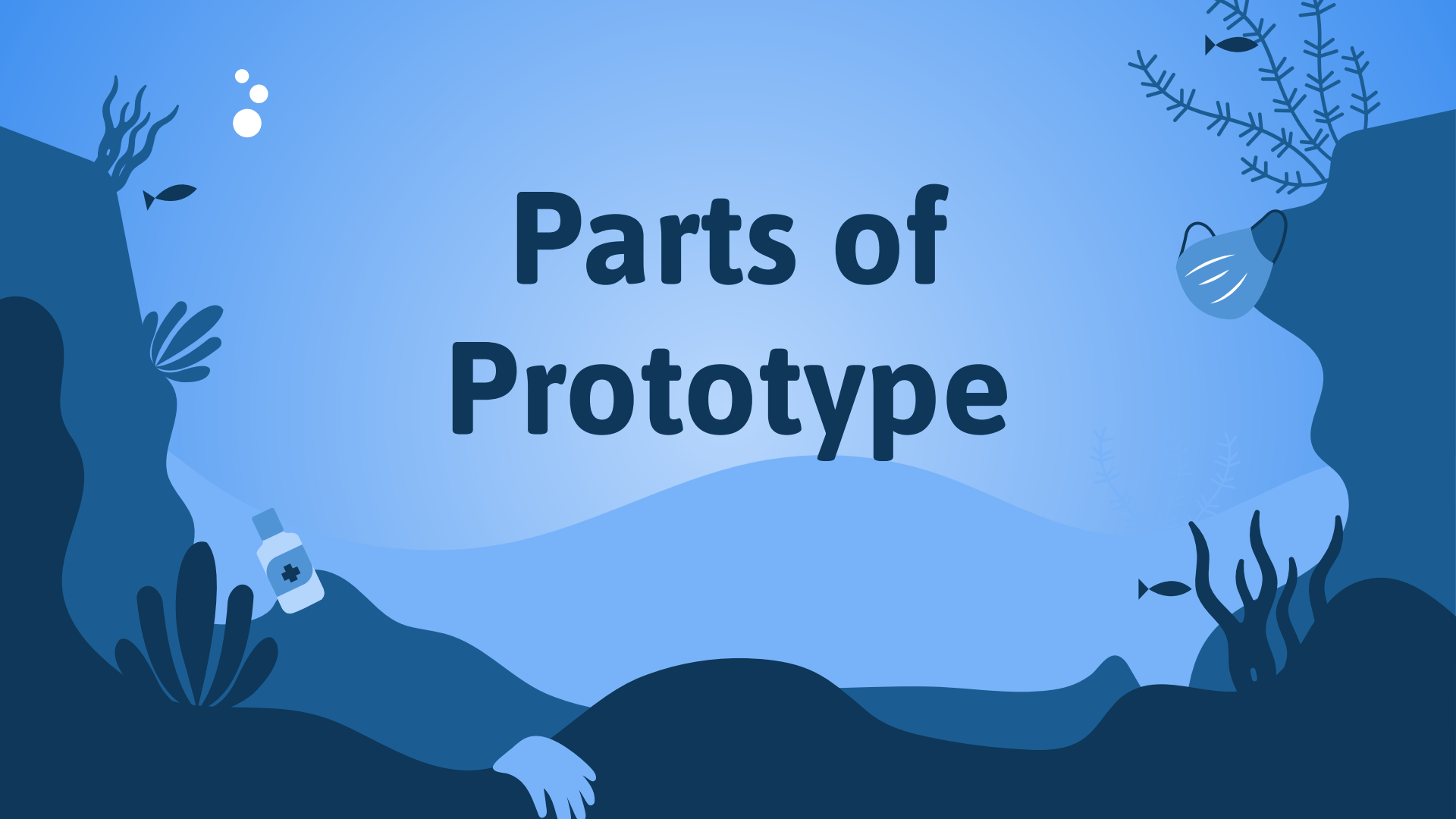
Assembly of Buoy



- Take 2 PVC pipes of diameter 3 inch
- Attach the two pipes using adhesive glue with a pipe connecting them
- Make apertures for the pipes and connect servos for opening and closing
- Make bottom pipe watertight
- Attach pipes to floats and attach 2 other floats , connected with hinges , micro servos for the storm mechanism
- Attach stirling engine with cassegrain apparatus and flettner rotor
- Connect the wheel of the stirling engine with bands to propellers at the bottom
- Connect and program the arduino, solar panels, load , tilt and acceleration sensors, GPS, etc.
- Fix in electrostatic precipitators



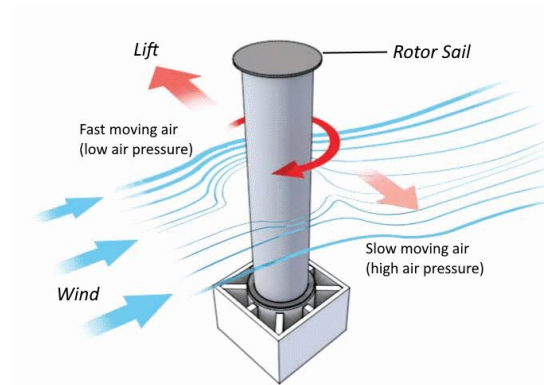
Parts of Prototype



<u>S.No.</u>	Component	Use
1	Stirling Engine	Powers the circuit board; Provides power to the propellers
2	Sails	Propels the machine
3	Circuit Board	Coordinates activities like the storm mechanism, sensors, sends capacity-full message
4	Solar Panels	Powers the circuit board
5	Set of Mirrors/ Lenses	Supply sufficient heat to stirling engine by concentrating heat
6	Flettner Rotor	Propels the machine
7	Thermos Insulation	Maintains constant temperature- colder than the upper plate of the Stirling engine, providing temperature differential; minimizes heat losses
8	Motor	Kick Starts the Stirling Engine, Draws in cables in the Storm Protection Mechanism, Turns sieves around when they are full



Stirling Engine with set of Mirrors

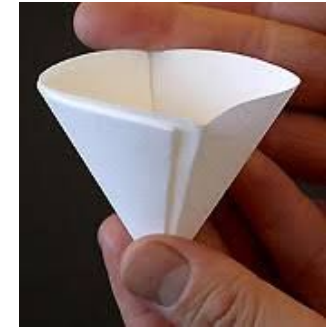


Flettner Rotor

S.No.	Component	Use
9	Greenhouse case	Using the greenhouse effect, heats up the top plate
10	Torque Promoter	Helps flywheel retain more torque
11	Primary Floats	keep the buoy afloat
12	Secondary Floats	part of storm mechanism- serve as shields against storms
13	Aperture	Closes the entrance and exit during storms
14	Coarse Sieve	Filters marine debris larger than 5mm, coarsely filters the particles
15	Marine Life Exit	outlet marine life escape so that marine life is unaffected by the buoy
16	Filters	filter out wastes and protects the internal components to increase lifespan
17	Fine Sieve	filters microparticles



Floats



Fine sieve



Coarse Sieve



Stirling engine with torque promoter and solar panels

<u>S.No.</u>	Component	Use
18	Tesla Valve	prevents backflow of water containing wastes during night when temperature differential is reversed, during storms, and prevents wastes from reaching the propellers, reduces back-currents
19	Propellers	propel the buoy and suck in the wastes
20	Converging Diverging Nozzle	Increase velocity of water, help in contributing speed and efficiency, constrict the space for passage of microparticles
21	Tanks	2 tanks filter together acting as backups to each other, dump wastes in third tank that ac
22	Electrostatic Precipitator	nuclear, metallic, volatile, and some chemical wastes filtered
23	Ultrasound sensor	senses when coarse sieves in both tanks are full
24	Float sensor	senses when tank is full by detecting the "sinkage"
25	Hall & tilt sensors	detects when there is a storm



Tesla Valve



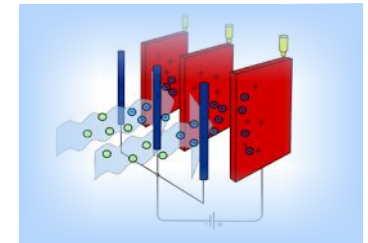
Float Sensor



Tilt Sensor



Ultrasound sensor

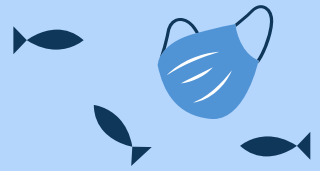


Electrostatic Precipitator

Tech used



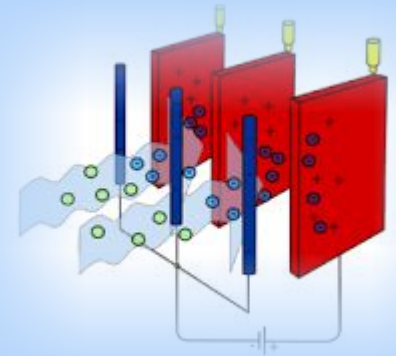
Tech used



Baleen whale



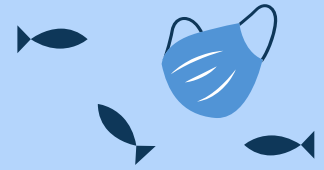
- Stirling Engine
- Flettner Rotor
- Tesla Valve
- Concentrated Solar Energy
- Greenhouse Effect
- Electrostatic Precipitator
- Biomimicry Of Baleen Structures Of Whales
- Torque Promoter
- Converging Diverging Nozzle
- Catamaran-type streamlining
- Cassegrain Apparatus



The background is a stylized underwater scene in various shades of blue. It features dark blue silhouettes of coral reefs and sea anemones on the left and right sides. Small white bubbles float in the upper left. A small white medical bottle with a blue cross is on the left, and a blue surgical mask is on the right. A hand is visible at the bottom center. Several small black fish are scattered throughout the scene.

How Uniquely It Solves the Problem

Unique Selling Proposition



- Revolutionary tech
- Efficient
- Eco-friendly
- Reduces landfills
- Cheap
- Large working radius
- Minimal energy wastage
- Faster cleanup in larger quantities
- Microplastic, chemical waste, and nuclear waste cleanup
- Active-passive hybrid, semi-autonomous



Cost Breakdown



Pipes-200

DIY parts-1000

float sensor -100

carpented parts-2000

Stirling engine-2700

Robotic parts:-

Solar panels(2)-450

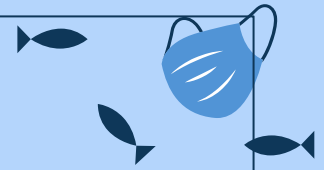
Micro servos(2)-260

Load cell-200

float sensor-100

Arduino Mega-970

Total-8590 INR*

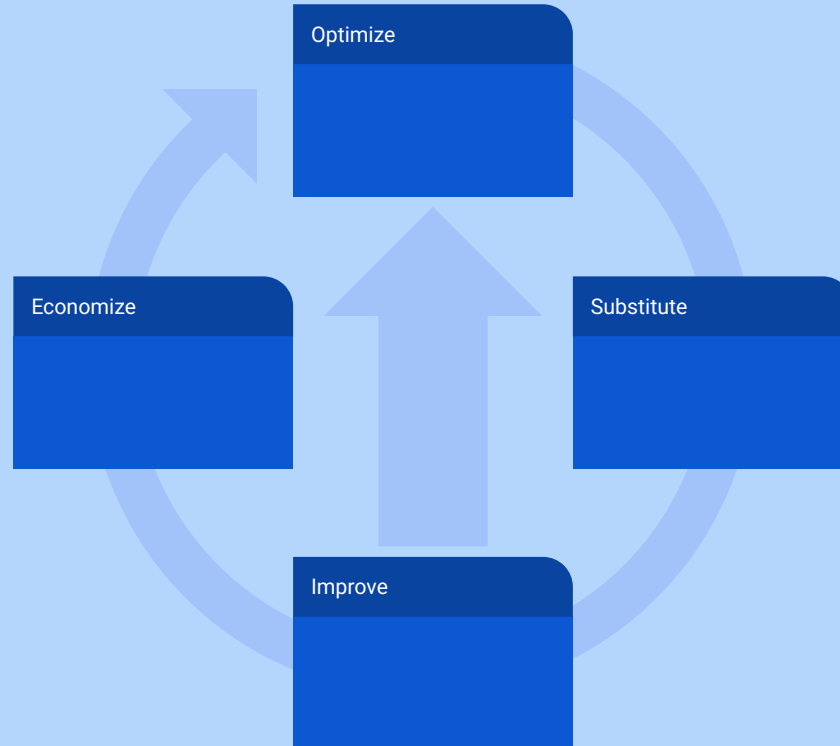
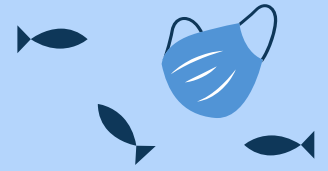


*Costs are far decreased when scrapyard material is used instead, as for the industry-grade machine

Changes Needed to Scale-up



Changes needed to scale-up



An underwater-themed illustration with a light blue background and dark blue silhouettes of rocks and coral. In the top left, there are three white bubbles of increasing size. A small black fish swims near the top left. On the right, a blue surgical mask floats near a piece of coral. In the bottom left, a white pill bottle with a blue cross symbol sits on a rock. In the bottom center, a white hand reaches up from the dark blue seabed. Another small black fish is visible near the bottom right.

Customers

Customers

The background is a light blue gradient representing water. It features several stylized blue fish swimming in different directions. There are clusters of white and yellow bubbles. At the bottom, there are dark blue wavy lines representing the ocean floor, with some light blue coral-like structures on the left side.

B2B

Ecologically conscious
Organizations, Reusing and
Recycling Organizations,
Decontaminators

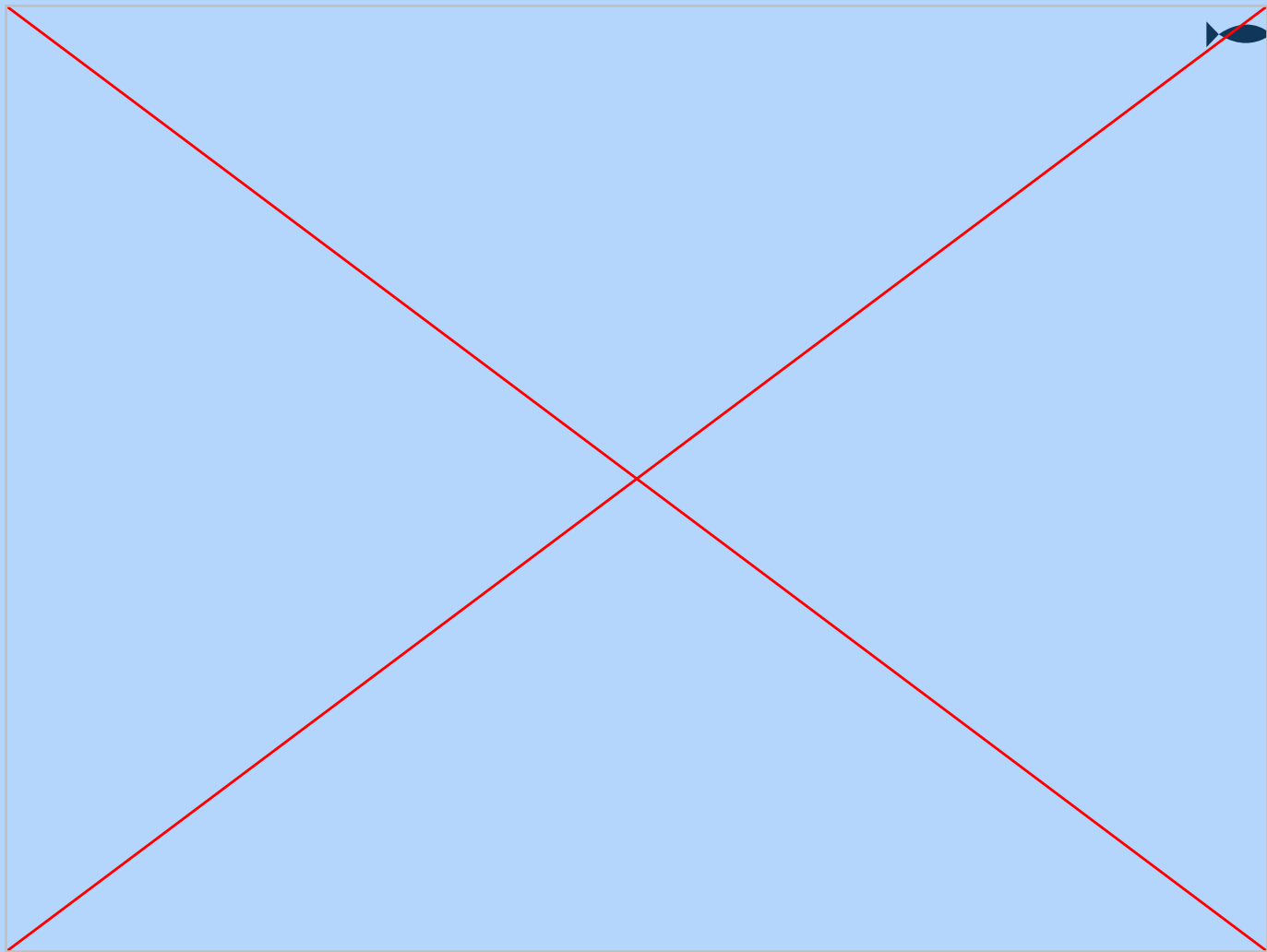
B2C

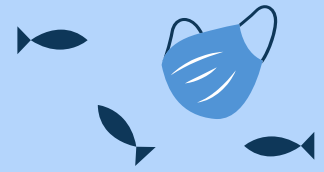
Individual Marine Life Activists,
Volunteers, Recyclers, Retailers

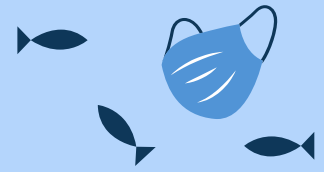
Scope

An underwater-themed illustration in various shades of blue. The word "Scope" is centered in a bold, dark blue font. The background features stylized coral reefs, small fish, bubbles, and medical-related icons: a white pill bottle with a blue cross on the left and a blue surgical mask on the right. A hand is visible at the bottom center, reaching upwards.



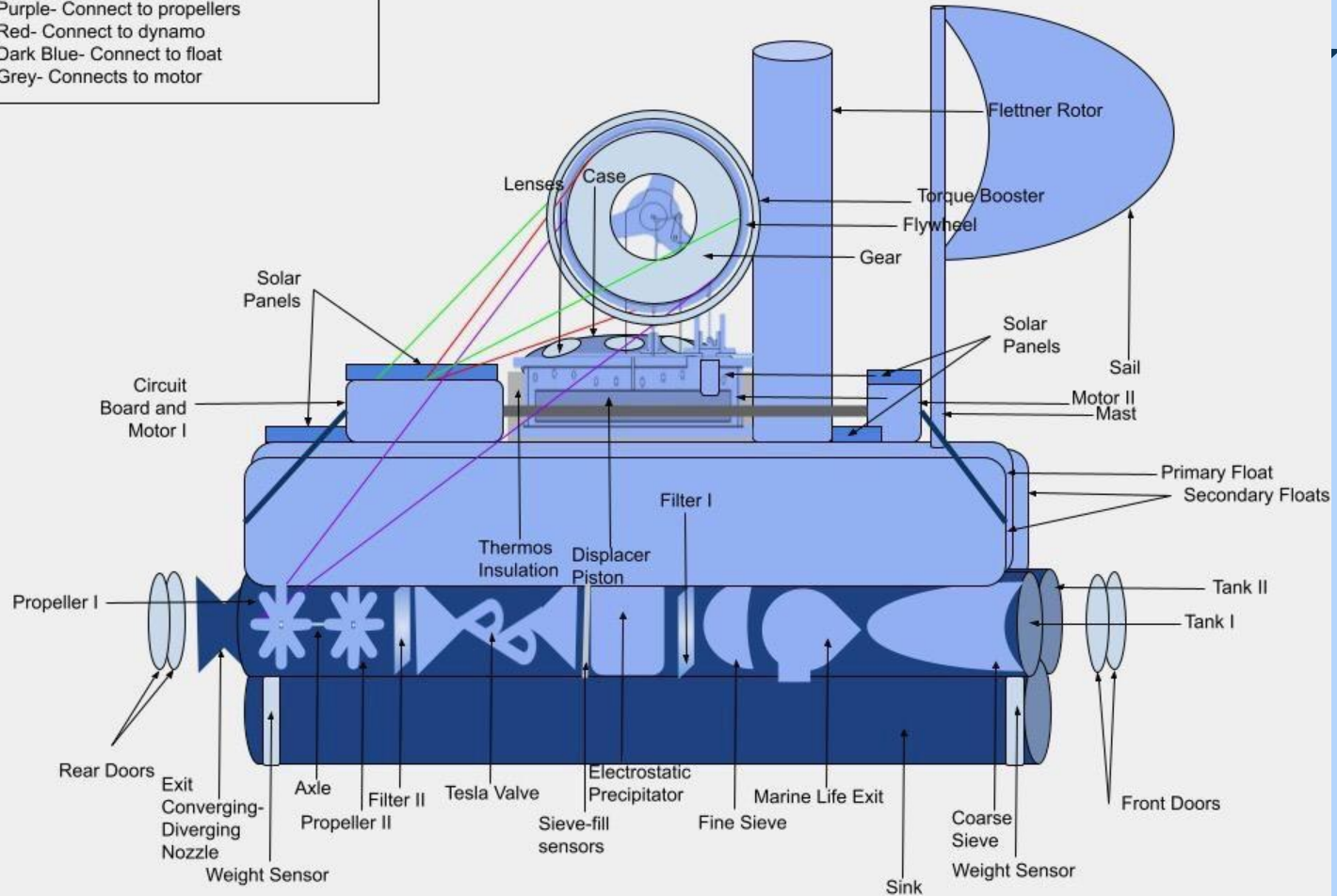






Legend for Colouring of Lines

Green- Connect to motor
 Purple- Connect to propellers
 Red- Connect to dynamo
 Dark Blue- Connect to float
 Grey- Connects to motor





Thankyou!

“The sea cures all ailments of man”

- Plato

